

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459128.7733 +/- 0.0058 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.006 +/- 0.062
Transit depth (Rp/Rs)^2	0.0042 +/- 0.08 [%]
Orbital Inclination (inc)	81.84 +/- 2.85
Airmass coefficient 1 (a1)	0.3 +/- 0.2
Airmass coefficient 2 (a2)	1.04 +/- 1.36
Scatter in the residuals of the lightcurve fit is	64.38 %
Best Comparison Star	None
Optimal Aperture	1.09
Optimal Annulus	3.32
Transit Duration (day)	0.024 +/- 0.028

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R fit vs published	0.006 ± 0.062 vs 0.1646 (deviation 2.56σ)
Duration fit vs published	0.024 d vs 0.0758 d (deviation 1.85σ)
Mid-time O-C	12.7 min
Depth SNR	0.1
Residual scatter	64.38 %

Light curve

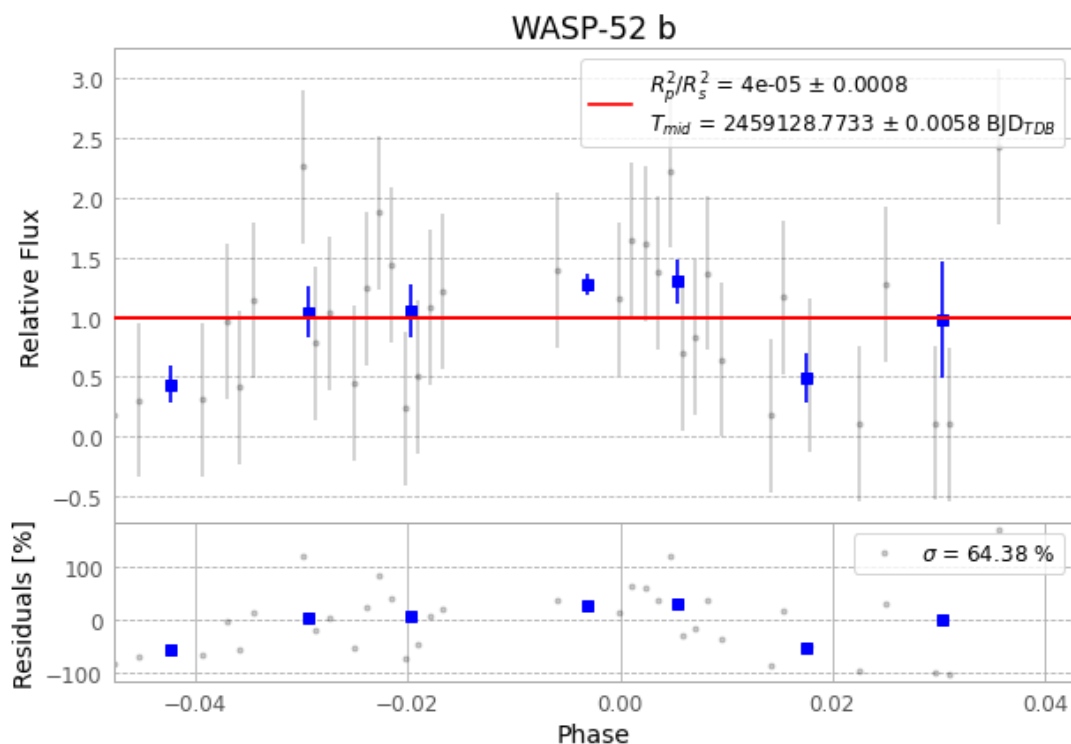


Figure 1 — FinalLightCurve_WASP-52 b_2020-10-05.png (SHA-256 4fc34d4a27435646...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [209, 337], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 9bdd2d1e630feccd...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

75 FITS frames (49 MB), WASP-52201006042412.FITS → WASP-52201006081212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-201006.log (d4d4884c1ad2...), FinalLightCurve_WASP-52 b_2020-10-05.png (4fc34d4a2743...), FinalLightCurve_WASP-52 b_2020-10-05.csv (9d82949dda46...)

Compute resources

Wall time 120.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 01:27Z.